# Deal memo: GB CCGT tolling agreement, calendar 2027

Status: DRAFT, ILLUSTRATIVE. GB power prices in this build are SIMULATED (no Ember file supplied) and gas is a Henry Hub-based proxy, so levels below demonstrate the method and are not a market view.

## Recommendation

On these inputs the toll is worth GBP 53,235,269 with start costs and a two-day minimum run, GBP 53,433,576 unconstrained, and GBP 42,361,787 intrinsic. The physical constraints cost GBP 198,307. The large gap between intrinsic and total value is the optionality of running only when the spark spread is positive, about 239 days a year here. The size of the value reflects the simulated spread (power forward average GBP 81.1/MWh against gas GBP 21.3/MWh), so the figure should not be quoted until real GB prices replace the simulation.

## Term sheet

| Item | Value |
|---|---|
| **Capacity** | 400 MW, 365 days |
| **Heat rate** | 2.0 MWh gas per MWh power |
| **Variable O&M** | GBP 4.0/MWh |
| **Carbon** | 0.37 tCO2/MWh at GBP 60.0/t (fixed input) |
| **Start cost / minimum run** | GBP 15,000 / 2 days |

## Valuation and cross-checks

| Method | GBP |
|---|---|
| **Kirk spread-option strip** | 51,301,665 |
| **Monte Carlo strip, no jumps** | 51,265,506 (SE 160,069) |
| **Monte Carlo strip, with jumps** | 53,433,576 |
| **Constrained dispatch** | 53,235,269 |

Kirk agrees with Monte Carlo within 1% once the correlation is converted to its terminal value under mean reversion. With no start cost and no minimum run the dispatch value equals the strip exactly.

## Sensitivities

| Input | Value GBP |
|---|---|
| **Correlation 0.3** | 53,043,691 |
| **Correlation 0.6** | 51,301,665 |
| **Correlation 0.8** | 50,086,836 |
| **Heat rate 1.8** | 62,162,500 |
| **Heat rate 2.0** | 51,301,665 |
| **Heat rate 2.2** | 41,693,335 |

Higher correlation lowers the value, because the spread becomes less volatile; the correlation of 0.6 is an assumption, since the estimated power-gas residual correlation from the simulated series is not informative.

## Hedge and limits

- Hedge the delta with power and gas forwards: about 2,597,554 MWh of power sold and 2,973,041 MWh of gas bought, rebalanced as spreads move.
- Price jumps are not hedgeable with forwards; the unhedged jump risk is shown in the hedging study (hedge.py).
- Carbon is a fixed price input; a carbon price series would add a third factor.
- No credit, ramp-rate or outage modelling.